

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25 Total Marks: 50 Annexure A

Code of Conduct:

I K.Neeha(192425245)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Neeha

Signature of the student:

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

1. Complete Workflow of a Model-Based Reinforcement Learning Agent

Model-Based Reinforcement Learning is an approach in which the agent learns an internal model of the environment and uses that model to predict future outcomes. The complete workflow begins with interaction between the agent and the environment.

Environment → State Observation → Action Selection → Reward & Next State → Experience Collection → Model Learning → Future Prediction → Planning → Policy Optimization → Optimal Action → Environment

The agent first observes the current state of the environment. It selects an action and receives a reward along with the next state. This experience is stored and used to learn the environment model. The model represents how the environment changes when an action is performed.

Once the model is learned, the planning component simulates different possible actions. It predicts their future states and rewards and identifies which action is likely to provide the highest long-term reward. The policy is then optimized using these planning results.

The improved policy selects the next action, which is executed in the actual environment. The new experience is again collected and used to update the model. Therefore, Model-Based RL follows a continuous **learning–planning–acting cycle**.

Workflow

- 1. Observe:** The agent observes the current state.
 - 2. Act:** It selects an action based on its current policy.
 - 3. Receive Feedback:** The environment returns a reward and next state.
 - 4. Learn Model:** The agent learns transition and reward patterns.
 - 5. Predict:** The model predicts possible future outcomes.
 - 6. Plan:** Different action sequences are evaluated.
 - 7. Optimize Policy:** The policy is improved according to predicted rewards.
 - 8. Execute:** The best action is performed in the real environment.
 - 9. Repeat:** The process continues until the desired behavior is learned.
- The major advantage of this workflow is that the agent can learn not only from real interactions but also from **simulated experiences generated by its learned model**. This can reduce the amount of costly interaction required with the real environment.

2. Architecture and Working of a Model-Based Reinforcement Learning System

A Model-Based RL system consists of several interconnected components that work together to enable learning, prediction, planning, and decision-making.

Environment → Experience → Model Learning → Internal Model → Planning → Policy Improvement → Action Execution → Environment

The **environment** is the real-world system in which the agent operates. It provides the current

state and responds to the actions performed by the agent.

The **experience collection component** records the interaction in the form: **(State, Action, Reward, Next State)**

These experiences are provided to the **model-learning component**. The model-learning component tries to understand the behavior of the environment. It learns two important aspects: how the environment changes after an action and what reward is expected from that action.

The learned information forms an **internal environment model**. This model allows the agent to simulate possible actions without actually performing every action in the real environment.

The **planning module** uses these predictions to compare alternative actions. It can perform simulations, rollouts, or search over possible future states. Based on the predicted rewards, it determines which action is more beneficial.

The **policy improvement component** updates the agent's decision-making strategy. A better policy assigns higher preference to actions that are expected to generate higher long-term rewards.

Finally, the **execution component** performs the selected action in the actual environment. The resulting state and reward are collected as new experience, which is used to further improve the model.

Main Components

Environment: Provides states, transitions, and rewards.

Experience Buffer: Stores previous interactions.

Model Learner: Learns environmental dynamics.

Environment Model: Predicts next states and rewards.

Planning Module: Simulates future possibilities.

Policy: Determines which action should be selected.

Execution Module: Performs the selected action.

Working Principle

The architecture can be summarized as:

**Real Experience → Model Learning → Simulation → Planning → Policy Improvement
→ Real Action → New Experience**

This creates a feedback loop. As more experience is collected, the model becomes more accurate. A better model allows better planning, and better planning leads to an improved policy.

This approach is particularly useful when interaction with the real environment is expensive. For example, in robotics, repeatedly testing unsafe actions on a physical robot can be costly or dangerous. A learned model can allow many actions to be tested virtually before the safest action is executed.

3. Model-Based RL vs Model-Free RL – Conceptual Comparison

Reinforcement Learning methods can broadly be divided into **Model-Based** and **Model-Free** approaches. The major difference is whether the agent learns and uses an explicit model of the environment.

Model-Based Reinforcement Learning

Environment Interaction → Experience → Environment Model → Prediction → Planning → Policy → Action

In Model-Based RL, the agent attempts to learn how the environment behaves. It estimates what next state and reward are likely to result from a particular action.

For example, a delivery robot can learn that moving through a particular road under certain traffic conditions may increase travel time. It can then use this knowledge to compare several routes before selecting one.

The important characteristic is **planning**. The agent can mentally simulate different future situations and select an action based on these predictions.

Model-Free Reinforcement Learning

Environment Interaction → Experience → Value/Policy Learning → Action

In Model-Free RL, the agent does not explicitly construct a model of the environment. Instead, it learns directly from experience which actions are useful.

For example, a delivery robot may repeatedly try different routes and learn that one particular route usually provides higher rewards. It learns the preferred action directly rather than building a detailed model of traffic and road transitions.

Comparison

Aspect	Model-Based RL	Model-Free RL
Environment model	Learned or available	Not explicitly required
Main idea	Predict and plan	Learn directly from experience
Planning	Important component	Usually not explicit
Data efficiency	Generally higher	Generally lower
Computational requirement	Higher because of simulation/planning	Usually lower during decision making
Adaptability	Depends on model accuracy	Learns directly from experience
Uncertainty	Can reason about predicted outcomes	Learns uncertainty through repeated experience

Aspect	Model-Based RL	Model-Free RL
Learning speed	Can be faster with a good model	
Main risk	Incorrect model predictions	Slow learning from limited experience
Examples	Dyna-Q, MCTS-based approaches	Q-Learning, SARSA, DQN

Relationship

Both approaches follow the basic RL principle of maximizing cumulative reward, but they learn in different ways.

Model-Based RL:

Learn Environment → Predict Future → Plan → Select Action

Model-Free RL:

Experience → Learn Action Value/Policy → Select Action

Advantages of Model-Based RL

Model-Based RL can be highly **sample-efficient** because the learned model can generate simulated experiences. The agent can evaluate several possible futures without interacting with the real environment each time. It is useful when real-world experimentation is expensive.

However, an inaccurate model can produce incorrect predictions. If the model does not represent the real environment properly, planning based on that model can lead to poor decisions.

Advantages of Model-Free RL

Model-Free RL is comparatively straightforward because it does not require construction of an accurate environment model. It can directly learn effective behavior from interaction and can perform well in environments that are difficult to model mathematically.

Its major limitation is that it can require a large number of interactions before learning good behavior.

Final Concept

The easiest way to remember the difference is:

Model-Free RL = “What action has worked well before?”

Therefore, Model-Based RL is more focused on **prediction and planning**, whereas Model Free RL is focused on **direct learning from experience**. Both approaches are important, and the choice depends on the complexity of the environment, availability of data, computational resources, and cost of real-world interaction.

RUBRICS FOR CALCULATION

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Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in depth, and insightful technical explanation	Technically correct content with adequate depth and	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood

			clarity		
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly	Appropriate tools and data used effectively	Limited use of tools and	Minimal or incorrect use of tools and

		effective visuals	with clear visuals	basic data interpretation	data; poor visuals
Communication	15	Highly engaging, confident, and audience focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

